

FATIMA MAMU

fatimaalkalimamu@gmail.com | 863-317-5742 | [GitHub](#) | [Personal Web Profile](#) | [LinkedIn](#)

San Francisco, CA, 94105 - Happy to relocate

PROFESSIONAL SUMMARY

Engineer with over a decade of professional experience building and shipping user-facing systems across real-time workflows. Spanning full-stack development, user research, and end-to-end interaction architecture. Has deployed voice AI systems in live clinical environments and brings firsthand operational experience running a café, a cake production studio, and a private jet catering service.

EDUCATION

Harvard University, Cambridge, USA — Master in Design Engineering

University College London, London, UK — Diploma in Computer Graphics, Vision and Imaging

Brunel University, London, UK — BEng, Electrical and Electronics Engineering

TECHNICAL SKILLS

Engineering & Infrastructure: REST APIs · Webhooks · WebSockets · Node.js · Python · TypeScript · Google Cloud · PostgreSQL · Supabase

Frontend & CMS: React · Next.js · TypeScript · Tailwind CSS · Webflow · Figma · design systems

Applied AI & Data (Engineering): Model training workflows · RAG pipelines · Vector search integration · Synthetic data generation

Voice & Conversational AI: Speech processing · Conversational AI agents · Real-time audio pipelines · Structured data extraction from voice · Twilio

WORK EXPERIENCE

AI Product Engineer | Lexi AI, Boston, MA, USA | 2025 – Present

- Improved clinician workflow efficiency by approximately 40% across four healthcare facilities by designing and deploying real-time interpretation features used in 7,000+ patient conversations
- Built a voice-based AI intake agent that collects patient information over the phone, auto-populates pre-arrival forms, and delivers a verification link via SMS, reducing manual form-filling burden for older patients less comfortable with online interfaces.
- Built real-time voice interaction workflows integrating speech processing, structured data capture, and workflow-aware interfaces to support clinical intake and interpretation services.
- Conducted on-site user research with clinicians across pilot facilities, translating workflow observations into targeted improvements that reduced friction in real clinical environments.
- Worked with product leadership and engineering to ship early product milestones that supported the company's \$1.4M seed round.

Creative Software Development Engineer - Summer | Walt Disney Imagineering (R & D), Glendale, CA, USA | 2024

- Developed real-time interactive installations using Unity, Maya, and TouchDesigner, including motion-driven water simulations for the Moana Journey of Water exhibit showcased at D23.
- Prototyped a glasses-free 3D display system for theme park attractions, working across animated scene creation, hand-tracking integration, and performance evaluation.
- Assisted in synchronizing TouchDesigner animation outputs with LED stage displays during physical staging and calibration of robotics-driven illusion systems involving Kuka robots.

Engineering Associate (Software & Systems Prototyping) | The Business People Limited | 2019 – 2023

- Designed and developed client-facing web platforms across multiple industries using Next.js, TypeScript, and PostgreSQL, supporting hundreds of active users.
- Conducted on-site workflow analysis across agriculture, aviation, and healthcare sectors to identify automation opportunities and prototype hardware and software solutions.

Founder & Creative Director | Edible Art Studio, Nigeria | 2013 – 2023

- Founded and operated a bespoke cake design studio for 10 years, managing end-to-end client relationships, creative briefs, production timelines, and a team of 15 across thousands of commissioned projects.
- Built operational systems for client intake, custom order workflows, and supplier coordination, scaling the business sustainably over a decade.

SELECTED PROJECTS

UMMA NA: Geo-data-based Emergency Maternal Health Platform | Full-Stack Development & UX Research | 2024 [See more](#)

- Bridged the gap between healthcare data and UX by translating complex maternal health protocols into a simplified mobile interface, reducing the time to connect patients with specialized care by 30% in pilot tests.
- Architected a real-time coordination system using **React** and **Firebase NoSQL streams**. Developed a custom routing algorithm using the **Google Maps Matrix API** to calculate condition-aware transport times for high-risk obstetric emergencies.

Soma: ML-based Biofeedback & EEG | Human-Computer Interaction (HCI) | 2023 [See more](#)

- Engineered a data pipeline to process live **EEG biosignals** using the **Muse SDK**. Developed a **PyTorch-based regression model** to map neural oscillations to generative visual parameters in **OpenCV**, creating a closed-loop biofeedback system.
- Explored the "trust in intelligent systems" by researching how users perceive agency when AI mediates their physiological data - a key focus for Microsoft's **human-centric AI** initiatives.

U-GO: AI Soccer Coaching Assistant | Computer Vision & Data Pipeline | 2024 - 2025 [See more](#)

- Conducted interviews and field observations with coaches across 3 cities; synthesised findings into interface requirements prioritising visual feedback over numerical metrics
- Built an end-to-end CV pipeline using **YOLOv8** for object detection and **MediaPipe** for skeletal pose estimation. Applied **Homography transformations** to project 2D camera coordinates into a 3D tactical bird's-eye view.
- Reached **80% detection accuracy** in uncontrolled outdoor lighting. The project serves as a prime example of building **ML-powered product experiences** that turn raw data into actionable user insights.